1 Law: slope only

($h := \bar{w} - \bar{w}_s$, $\bar{w}_s$ = visible surface, given; $g := h + \delta$ the shifted gap; $\delta > 0$ = effective wall below the visible surface; contact slope pinned: $\bar{a}'(g = 1) = 1$ for every p; δ, p = the two constants; used on $h \geq 1.1$)

$$\bar{a}'(h;\, \delta,\, p) = \frac{1}{(h+\delta)^p}$$

$$\frac{\partial \bar{a}'}{\partial h} = -\frac{p\,\bar{a}'}{h+\delta}$$

$$\frac{\partial \bar{a}'}{\partial \delta} = -\frac{p\,\bar{a}'}{h+\delta} \qquad = \frac{\partial \bar{a}'}{\partial h}$$

$$\frac{\partial \bar{a}'}{\partial p} = -\,\bar{a}'\,\ln(h+\delta)$$

$$\frac{\partial \bar{a}'}{\partial \bar{a}_w} = 0$$

(integrated once, for the seed container only, $\bar{a}(\infty) = 1$; the value law is NOT used by the estimator)

$$\int_h^\infty \bar{a}'\,dh' = \frac{1}{(p-1)(h+\delta)^{p-1}}$$

$$\bar{a}(h;\, \delta,\, p) = 1 - \frac{1}{(p-1)(h+\delta)^{p-1}}$$

$$\frac{\partial \bar{a}}{\partial \delta} = \frac{1}{(h+\delta)^p} = \bar{a}'$$

$$\frac{\partial \bar{a}}{\partial p} = \frac{\dfrac{1}{p-1} + \ln(h+\delta)}{(p-1)(h+\delta)^{p-1}}$$

(family residual on a wall with constants (δ_w, p_w), and the gain error it leaves along a descent from h_0 if never corrected)

$$r'(h) := \bar{a}'_{\text{true}}(h) - \bar{a}'(h;\, \delta_w,\, p_w)$$

$$E(h) := \int_h^{h_0} r'(h')\,dh'$$

2 State

$\mathbf{x} = \left[\, \delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ p,\ \delta \,\right]^\top$ (code slots: $5 = p$, $7 = \delta$, 6 locked)

3 True dynamics

($d = 2$; $\bar{a}'_{\text{true}}$ = the plant's slope)

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}\big(\bar{w}[k] - \bar{w}_s\big)\,\Delta\bar{w}[k] \\
p[k+1] &= p[k] \\
\delta[k+1] &= \delta[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]
\end{aligned}
$$

4 Estimator

$(\hat{\bar{w}}[k] = \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k];\ \hat{h}[k] = \hat{\bar{w}}[k] - \bar{w}_s;\ \hat{g}[k] = \hat{h}[k] + \hat{\delta}[k];\ M[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k])$

$$
\begin{aligned}
\bar{a}'[k] &= \frac{1}{\hat{g}[k]^{\hat{p}[k]}} \\
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \bar{a}'[k]\,M[k] \\
\hat{p}[k+1] &= \hat{p}[k] \\
\hat{\delta}[k+1] &= \hat{\delta}[k]
\end{aligned}
$$

(exact step over the known displacement M, used in code; first order in M it is $\bar{a}'[k]\,M[k]$)

$$
\hat{\bar{a}}_w[k+1] = \hat{\bar{a}}_w[k] + \frac{\hat{g}[k]^{\,1-\hat{p}[k]} - \big(\hat{g}[k] + M[k]\big)^{1-\hat{p}[k]}}{\hat{p}[k] - 1}
$$

5 Error dynamics

($e = \text{true} - \text{estimate}$; first order at the estimate; $\Delta\bar{w} - M = (1-\lambda_c)\,e_{\delta\bar{w}_3} + F_{dw}\,e_{\bar{a}_w} + \varepsilon_w$; $h - \hat{h} = -e_{\delta\bar{w}_3}$; $\delta - \hat{\delta} = e_\delta$; $p - \hat{p} = e_p$; $g - \hat{g} = e_\delta - e_{\delta\bar{w}_3}$)

$$\bar{a}'_{\text{true}} - \bar{a}'[k] = \frac{\hat{p}[k]\,\bar{a}'[k]}{\hat{g}[k]}\, e_{\delta\bar{w}_3}[k] - \frac{\hat{p}[k]\,\bar{a}'[k]}{\hat{g}[k]}\, e_\delta[k] - \bar{a}'[k] \ln \hat{g}[k]\; e_p[k] + r'[k]$$

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\, e_{\delta\bar{w}_3}[k] - F_{dw}[k]\, e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \Big(\bar{a}'[k](1-\lambda_c) + \frac{\hat{p}[k]\,\bar{a}'[k]}{\hat{g}[k]} M[k]\Big) e_{\delta\bar{w}_3}[k] + \big(1 + \bar{a}'[k] F_{dw}[k]\big) e_{\bar{a}_w}[k] \\
&\qquad - \bar{a}'[k] \ln \hat{g}[k]\, M[k]\, e_p[k] - \frac{\hat{p}[k]\,\bar{a}'[k]\, M[k]}{\hat{g}[k]}\, e_\delta[k] + \bar{a}'[k]\, \varepsilon_w[k] + M[k]\, r'[k] \\
e_p[k+1] &= e_p[k] \\
e_\delta[k+1] &= e_\delta[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dw}[k] & 0 & 0 \\
0 & 0 & \bar{a}'[k](1-\lambda_c) + \dfrac{\hat{p}[k]\,\bar{a}'[k]}{\hat{g}[k]} M[k] & 1 + \bar{a}'[k] F_{dw}[k] & -\bar{a}'[k] \ln \hat{g}[k]\, M[k] & -\dfrac{\hat{p}[k]\,\bar{a}'[k]\, M[k]}{\hat{g}[k]} \\
0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}$$

$$F_{e,46}[k] = -\frac{\hat{p}[k]\,\bar{a}'[k]}{\hat{g}[k]}\, M[k] \qquad = -\big(\text{the height share of } F_{e,43}\big) \qquad \text{(the slope sees } e_\delta - e_{\delta\bar{w}_3} \text{ only)}$$

6 Error dynamics, row 4 expanded

(all quantities at step k; e = true − estimate; $h = \hat{h} - e_{\delta\bar{w}_3}$; $\delta = \hat{\delta} + e_\delta$; $p = \hat{p} + e_p$)

(a) the two increments and their difference

$$\bar{a}_w[k+1] = \bar{a}_w[k] + \bar{a}'_{\text{true}}\,\Delta\bar{w}$$
$$\hat{\bar{a}}_w[k+1] = \hat{\bar{a}}_w[k] + \bar{a}'\,M$$
$$e_{\bar{a}_w}[k+1] = e_{\bar{a}_w}[k] + \bar{a}'_{\text{true}}\,\Delta\bar{w} - \bar{a}'\,M$$

(b) the two factors, each written as estimate plus its error

$$\Delta\bar{w} - M = (1-\lambda_c)\,e_{\delta\bar{w}_3} + F_{dw}\,e_{\bar{a}_w} + \varepsilon_w$$

$$\bar{a}'_{\text{true}}(h) = \bar{a}'(h;\,\delta,\,p) + r'$$

$$\bar{a}'(h;\,\delta,\,p) - \bar{a}'(\hat{h};\,\hat{\delta},\,\hat{p}) = \frac{\partial\bar{a}'}{\partial g}\left[(h+\delta) - (\hat{h}+\hat{\delta})\right] + \frac{\partial\bar{a}'}{\partial p}\,(p-\hat{p}) \qquad \text{first order}$$

$$= \left(-\frac{\hat{p}\,\bar{a}'}{\hat{g}}\right)(e_\delta - e_{\delta\bar{w}_3}) + \left(-\bar{a}'\ln\hat{g}\right)(e_p)$$

$$\bar{a}'_{\text{true}} - \bar{a}' = \frac{\hat{p}\,\bar{a}'}{\hat{g}}\,e_{\delta\bar{w}_3} - \frac{\hat{p}\,\bar{a}'}{\hat{g}}\,e_\delta - \bar{a}'\ln\hat{g}\;e_p + r'$$

(c) the product difference, exact

$$\bar{a}'_{\text{true}}\,\Delta\bar{w} - \bar{a}'\,M = (\bar{a}'_{\text{true}} - \bar{a}')\,M + \bar{a}'\,(\Delta\bar{w} - M) + (\bar{a}'_{\text{true}} - \bar{a}')(\Delta\bar{w} - M)$$

(d) every term after substituting (b)

$$
\begin{aligned}
(\bar{a}'_{\text{true}} - \bar{a}')\, M = {} & \frac{\hat{p}\,\bar{a}'}{\hat{g}}\, M\, e_{\delta\bar{w}_3} && (1)\\
& - \frac{\hat{p}\,\bar{a}'}{\hat{g}}\, M\, e_{\delta} && (2)\\
& - \bar{a}' \ln \hat{g}\, M\, e_p && (3)\\
& + M\, r' && (4)
\end{aligned}
$$

$$
\begin{aligned}
\bar{a}'\,(\Delta\bar{w} - M) = {} & \bar{a}'\,(1-\lambda_c)\, e_{\delta\bar{w}_3} && (5)\\
& + \bar{a}'\, F_{dw}\, e_{\bar{a}_w} && (6)\\
& + \bar{a}'\, \varepsilon_w && (7)
\end{aligned}
$$

$$
\begin{aligned}
(\bar{a}'_{\text{true}} - \bar{a}')(\Delta\bar{w} - M) = {} & \frac{\hat{p}\,\bar{a}'}{\hat{g}}\,(1-\lambda_c)\, e_{\delta\bar{w}_3}\, e_{\delta\bar{w}_3} && (8)\ \text{second order}\\
& + \frac{\hat{p}\,\bar{a}'}{\hat{g}}\, F_{dw}\, e_{\delta\bar{w}_3}\, e_{\bar{a}_w} && (9)\ \text{second order}\\
& + \frac{\hat{p}\,\bar{a}'}{\hat{g}}\, e_{\delta\bar{w}_3}\, \varepsilon_w && (10)\ \text{second order}\\
& - \frac{\hat{p}\,\bar{a}'}{\hat{g}}\,(1-\lambda_c)\, e_{\delta}\, e_{\delta\bar{w}_3} && (11)\ \text{second order}\\
& - \frac{\hat{p}\,\bar{a}'}{\hat{g}}\, F_{dw}\, e_{\delta}\, e_{\bar{a}_w} && (12)\ \text{second order}\\
& - \frac{\hat{p}\,\bar{a}'}{\hat{g}}\, e_{\delta}\, \varepsilon_w && (13)\ \text{second order}\\
& - \bar{a}' \ln \hat{g}\,(1-\lambda_c)\, e_p\, e_{\delta\bar{w}_3} && (14)\ \text{second order}\\
& - \bar{a}' \ln \hat{g}\, F_{dw}\, e_p\, e_{\bar{a}_w} && (15)\ \text{second order}\\
& - \bar{a}' \ln \hat{g}\, e_p\, \varepsilon_w && (16)\ \text{second order}\\
& + (1-\lambda_c)\, r'\, e_{\delta\bar{w}_3} && (17)\ \text{second order}\\
& + F_{dw}\, r'\, e_{\bar{a}_w} && (18)\ \text{second order}\\
& + r'\, \varepsilon_w && (19)\ \text{second order}
\end{aligned}
$$

(e) first-order terms collected by error

$$
\begin{aligned}
e_{\bar{a}_w}[k+1] &= e_{\bar{a}_w}[k] + (1) + (5) + (6) + (3) + (2) + (7) + (4)\\
&= \Big(\bar{a}'\,(1-\lambda_c) + \frac{\hat{p}\,\bar{a}'}{\hat{g}}\, M\Big)\, e_{\delta\bar{w}_3} && (5)+(1) \quad \to F_e(4,3)\\
&\quad + \big(1 + \bar{a}'\, F_{dw}\big)\, e_{\bar{a}_w} && 1+(6) \quad \to F_e(4,4)\\
&\quad - \bar{a}' \ln \hat{g}\, M\, e_p && (3) \quad \to F_e(4,5)\\
&\quad - \frac{\hat{p}\,\bar{a}'}{\hat{g}}\, M\, e_{\delta} && (2) \quad \to F_e(4,6)\\
&\quad + \bar{a}'\, \varepsilon_w && (7) \quad \to Q,\ \mathbf{g}_4 = \bar{a}'\\
&\quad + M\, r' && (4) \quad \to \text{not in } F_e,\ \text{not in } Q
\end{aligned}
$$

7 Measurement

$(\nabla_d \bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d])$

$$y_1[k] = \delta\bar{w}_1[k] + n_w[k]$$
$$y_2[k] = \bar{a}_w[k-d] + n_a[k]$$
$$\hat{y}_1[k] = \delta\hat{\bar{w}}_1[k]$$
$$\hat{y}_2[k] = \hat{\bar{a}}_w[k] - \bar{a}'[k]\,\nabla_d\bar{w}_d[k]$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\dfrac{\hat{p}[k]\,\bar{a}'[k]}{\hat{g}[k]}\nabla_d\bar{w}_d[k] & 1 & \bar{a}'[k]\ln\hat{g}[k]\,\nabla_d\bar{w}_d[k] & \dfrac{\hat{p}[k]\,\bar{a}'[k]}{\hat{g}[k]}\nabla_d\bar{w}_d[k] \end{bmatrix}$$

$H_{26}[k] = -\,H_{23}[k]$ (the walk-back sees $e_\delta - e_{\delta\bar{w}_3}$ only)

8 Noise

$$Q[k] = Q_{33}[k]\,\mathbf{g}\mathbf{g}^\top, \qquad \mathbf{g} = \big[\,0,\ 0,\ -1,\ \bar{a}'[k],\ 0,\ 0\,\big]^\top, \qquad Q_{55} = Q_{66} = 0$$
$$R[k] = \mathrm{diag}\big(R_1,\ R_2[k]\big)$$

9 Observability

$(\Phi(k,k_0) := F_e[k-1]\cdots F_e[k_0],\ \Phi(k_0,k_0) = I;\ N$ window length)

Gate 1, the two columns of each constant:

$$F_{e,45}[k] = -\,\bar{a}'[k]\ln\hat{g}[k]\,M[k], \qquad H_{25}[k] = \bar{a}'[k]\ln\hat{g}[k]\,\nabla_d\bar{w}_d[k]$$
$$F_{e,46}[k] = -\,\frac{\hat{p}[k]\,\bar{a}'[k]}{\hat{g}[k]}\,M[k], \qquad H_{26}[k] = \frac{\hat{p}[k]\,\bar{a}'[k]}{\hat{g}[k]}\,\nabla_d\bar{w}_d[k]$$

(all four proportional to a displacement: zero in a hold; all four $\to 0$ as $\hat{g} \to \infty$; the p columns vanish at $\hat{g} = 1$ exactly, the pinned contact, which is the band edge and not an interior point)

Gate 2, time signatures:

$e_p[k+1] = e_p[k]$, $\quad e_\delta[k+1] = e_\delta[k]$ (two constants)

$$e_{\bar{a}_w}[k+1] = \big(1 + \bar{a}'F_{dw}\big)e_{\bar{a}_w} - \bar{a}'M\Big(\ln\hat{g}\ e_p + \frac{\hat{p}}{\hat{g}}\,e_\delta\Big) + \ldots$$

(one level, drifting with the height travelled; at one height only the combination $\ln\hat{g}\ e_p + \frac{\hat{p}}{\hat{g}}e_\delta$ is seen)

$$e_{\delta\bar{w}_3}[k+1] = \lambda_c\,e_{\delta\bar{w}_3} - F_{dw}\,e_{\bar{a}_w} - \varepsilon_w, \qquad y_1[k+2] = e_{\delta\bar{w}_3}[k] + \ldots$$

($e_{\delta\bar{w}_3}$ has the λ_c dynamics and is read by y_1; e_δ is a constant: not an alias although $F_{e,46} = -$height share of $F_{e,43}$)

Gate 3, $\det O_2$ with coefficients frozen ($\alpha := 1 + \bar{a}' F_{dw}$, $q := -F_{dw}$; $\beta_p := \bar{a}' \ln \hat{g}\, M$, $\gamma_p := \bar{a}' \ln \hat{g}\, \nabla_d \bar{w}_d$; $\beta_\delta := \hat{p}\bar{a}' M/\hat{g}$, $\gamma_\delta := \hat{p}\bar{a}' \nabla_d \bar{w}_d/\hat{g}$):

$$\text{pair } (e_{\bar{a}_w}, e_p),\ \text{row } y_2: \quad \det \begin{bmatrix} 1 & \gamma_p \\ \alpha & \gamma_p - \beta_p \end{bmatrix} = -\beta_p - \gamma_p(\alpha - 1) = -\bar{a}' \ln \hat{g} \left(M + \nabla_d \bar{w}_d\, \bar{a}' F_{dw} \right)$$

$$\text{pair } (e_{\bar{a}_w}, e_p),\ \text{row } y_1: \quad \det \begin{bmatrix} q & 0 \\ q\alpha & -q\beta_p \end{bmatrix} = -q^2 \beta_p = -F_{dw}^2\, \bar{a}' \ln \hat{g}\, M$$

$$\text{pair } (e_{\bar{a}_w}, e_\delta),\ \text{row } y_2: \quad \det \begin{bmatrix} 1 & \gamma_\delta \\ \alpha & \gamma_\delta - \beta_\delta \end{bmatrix} = -\beta_\delta - \gamma_\delta(\alpha - 1) = -\frac{\hat{p}\, \bar{a}'}{\hat{g}} \left(M + \nabla_d \bar{w}_d\, \bar{a}' F_{dw} \right)$$

$$\text{pair } (e_{\bar{a}_w}, e_\delta),\ \text{row } y_1: \quad \det \begin{bmatrix} q & 0 \\ q\alpha & -q\beta_\delta \end{bmatrix} = -q^2 \beta_\delta = -\frac{\hat{p}\, \bar{a}'\, F_{dw}^2\, M}{\hat{g}}$$

$$\text{triple } (e_{\bar{a}_w}, e_p, e_\delta),\ \text{row } y_2,\ \text{coefficients frozen:} \quad A_3 = \begin{bmatrix} \alpha & -\beta_p & -\beta_\delta \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad C_3 = \begin{bmatrix} 1,\ \gamma_p,\ \gamma_\delta \end{bmatrix}$$

$$\det \begin{bmatrix} C_3 \\ C_3 A_3 \\ C_3 A_3^2 \end{bmatrix} = 0 \qquad \text{identically} \qquad \left(\frac{\beta_\delta}{\beta_p} = \frac{\gamma_\delta}{\gamma_p} = \frac{\hat{p}}{\hat{g} \ln \hat{g}} \right)$$

(the two constants separate only through the change of that ratio with height: visible combinations at two heights)

$$\mathbf{v}(\hat{g}) := \begin{bmatrix} \ln \hat{g},\ \dfrac{\hat{p}}{\hat{g}} \end{bmatrix}$$

$$\det \begin{bmatrix} \mathbf{v}(\hat{g}_1) \\ \mathbf{v}(\hat{g}_2) \end{bmatrix} = \frac{\hat{p} \left(\hat{g}_1 \ln \hat{g}_1 - \hat{g}_2 \ln \hat{g}_2 \right)}{\hat{g}_1\, \hat{g}_2}$$

$$\det O_2 = 0 \text{ (either pair, } y_2) \iff M = 0 \text{ and } \nabla_d \bar{w}_d = 0 \text{ (hold)} \ \text{ or } \ \hat{g} \to \infty \ \text{ or (pair } p \text{ only) } \hat{g} = 1$$

$$\det O_2 = 0 \text{ (either pair, } y_1) \iff F_{dw} = 0 \text{ or } M = 0 \text{ (hold)} \ \text{ or } \ \hat{g} \to \infty \ \text{ or (pair } p \text{ only) } \hat{g} = 1$$

$$\det[\mathbf{v}(\hat{g}_1);\ \mathbf{v}(\hat{g}_2)] = 0 \iff \hat{g}_1 = \hat{g}_2 \text{ (one height; } x \ln x \text{ is monotone for } x > 1/e) \ \text{ or } \ \hat{g} \to \infty$$

no interior zero on the band: $\dfrac{\partial \bar{a}'}{\partial p} = -\bar{a}' \ln \hat{g} \neq 0$ for $\hat{g} > 1$, $\quad \dfrac{\partial \bar{a}'}{\partial \delta} = -\hat{p}\bar{a}'/\hat{g} \neq 0$

alias with the given surface: only $\bar{w}_s - \delta$ enters g; $\bar{w}_s$ is given, not a state

alias with $\delta \bar{w}_3$ in the slope: only $e_\delta - e_{\delta \bar{w}_3}$ enters; separated by gate 2 (dynamics and y_1)

compared with (δ, c): the second constant's level sensitivity

$\partial \bar{a}/\partial p = \frac{1 + (p-1)\ln g}{(p-1)^2 g^{p-1}}$ carries $(1 + \ln g)$ more than $\partial \bar{a}/\partial c = 1/g$ (probe_ident_map_f2.m)

Seed and prior

fit of the slope on the band $h \in [1.10,\ 22.2]$, least squares on $\bar{a}'$, contact slope pinned at 1 :

$(\delta,\ p)_{\text{plane}} = (+0.059,\ 1.950)$, $(\delta,\ p)_{\text{cell}} = (+1.115,\ 1.938)$

$(\delta,\ p)_{\text{sphere}} = (-0.035,\ 2.018)$ (plane-tail plant, sphere_cperp_fig11_ptail)

$\sup|r'|$ on the band: plane 0.022, cell 0.001, sphere 0.015

$\max_h |E(h)|$, $h_0 = 22.2$: plane 0.004, cell 0.004, sphere 0.025 ((δ, c): 0.012/0.001/0.025)

segments near/mid/far: δ_{plane} $+0.05/+0.08/-0.05$, p_{plane} 2.00/1.93/1.96

δ_{cell} $+1.14/+1.10/+0.97$, p_{cell} 1.92/1.94/1.96

protocol (plane init):

$\hat{\delta}[0] = \delta_{\text{plane}} = +0.059$

$\hat{p}[0] = p_{\text{plane}} = 1.950$

$\hat{\bar{a}}_w[0] = D_{\perp}\Big(\dfrac{1}{\bar{w}[0] - \bar{w}_s}\Big)$ (plane polynomial at the start height)

$\sigma_{\delta,0} = 0.64$ (kept equal to the (δ, c) runs for the paired comparison;
the family rule $\frac{1}{2}(\delta_{\text{cell}} - \delta_{\text{sphere}})$ gives 0.58)

$\sigma_{p,0} = 0.10$ (wall family $1.94 \ldots 2.02$; in-wall walk of p per segment 0.04–0.14)

$$P[0] = \begin{bmatrix} P_{11}[0] & & & & & \\ & P_{22}[0] & & & & \\ & & P_{33}[0] & & & \\ & & & P_{44}[0] & P_{45}[0] & P_{46}[0] \\ & & & P_{45}[0] & \sigma_{p,0}^2 & \\ & & & P_{46}[0] & & \sigma_{\delta,0}^2 \end{bmatrix}$$

$$P_{44}[0] = \Big(\frac{\partial \bar{a}}{\partial \delta}\Big)^2 \sigma_{\delta,0}^2 + \Big(\frac{\partial \bar{a}}{\partial p}\Big)^2 \sigma_{p,0}^2 + P_{a,\text{floor}}$$

$$= \bar{a}'(\hat{g}[0])^2\, \sigma_{\delta,0}^2 + \Big(\frac{\frac{1}{\hat{p}-1} + \ln \hat{g}[0]}{(\hat{p}-1)\, \hat{g}[0]^{\hat{p}-1}}\Big)^2 \sigma_{p,0}^2 + P_{a,\text{floor}}$$

$$P_{46}[0] = \frac{\partial \bar{a}}{\partial \delta}\, \sigma_{\delta,0}^2 = \bar{a}'(\hat{g}[0])\, \sigma_{\delta,0}^2$$

$$P_{45}[0] = \frac{\partial \bar{a}}{\partial p}\, \sigma_{p,0}^2$$

(the start value was read from the plane; the wall's (δ, p) move it by $\partial \bar{a}/\partial \theta$:
position-type container, used once, at $k = 0$; at 22.2 R : $\partial \bar{a}/\partial p = 0.23$, so $\sqrt{P_{44}[0]} \approx 0.023$)

$Q_{55} = Q_{66} = 0$

guard:

$\hat{g}[k] \geq g_{\min}$ (numerical floor below the band; the slope has no singularity for $\hat{g} > 0$)

$\hat{p}[k] \in [1.2,\ 4]$ (the offline search range; $p > 1$ for the exact step), band $\hat{h} \geq 1.1$

pre-registered pass (wall-test protocol, canon deep, 10 seeds, $e_{\bar{a}} = \bar{a}_{\text{true}} - \hat{\bar{a}}_w$ absolute), paired with (δ, c):

cell descent worst ≤ 0.05, hold $|e_{\bar{a}}| \leq 0.02$ on plane, cell, sphere;

cell $\hat{\delta}$ closer to its fit than under (δ, c) (1.08 of 1.24); $\hat{p}$ honesty 0.7–1.4;

plane and sphere descent worst not worse than (δ, c)